

PDD_STAT — User Guide

Version: 1.0

Developer: P. Demeshko

Interface language: English (labels, metrics, plot annotations)

Table of Contents

1. [System Overview](#)
2. [Installation & Startup](#)
3. [Project & Data Management](#)
4. 3.1. Creating a project and uploading data
5. 3.2. Data cleaning
6. 3.3. Variable management
7. 3.4. Project context (description, aim)
8. [Interface](#)
9. 4.1. Left panel — Project Manager
10. 4.2. Workspace
11. 4.3. Right panel — Variables
12. [Statistical Analysis Templates](#)
13. 5.1. Categorical Comparison
14. 5.2. Numeric Comparison
15. 5.3. Correlation Analysis
16. 5.4. Spline Analysis
17. 5.5. Descriptive Statistics
18. 5.6. Violin Plots
19. 5.7. ANOVA / Kruskal-Wallis
20. 5.8. Chart Builder
21. 5.9. Cox Regression
22. 5.10. Logistic Regression
23. 5.11. Logistic LASSO
24. 5.12. Random Forest
25. 5.13. Kaplan-Meier
26. 5.14. ROC Analysis
27. 5.15. Model Evaluation (Binary)
28. 5.16. Survival Prediction Evaluation

- 29. 5.17. Diagnostic Accuracy
 - 30. 5.18. Agreement Analysis
 - 31. 5.19. Individual Prediction
 - 32. [AI Chat](#)
 - 33. 6.1. AI Settings
 - 34. 6.2. Modes: Assistant and Coder
 - 35. [Reports](#)
 - 36. 7.1. Standard DOCX Report
 - 37. 7.2. AI Report
 - 38. [Code Editor Mode](#)
 - 39. [Analysis History \(Chain\)](#)
 - 40. [Tips & Best Practices](#)
-

1. System Overview

PDD_STAT is a web application for statistical analysis of medical and clinical data. The system provides:

- **22 analysis templates** (from descriptive statistics to regression models and survival analysis)
- **AI assistant** powered by Ollama (local) or Groq (cloud API)
- **DOCX report generation** (standard and AI-generated)
- **Code editor** for custom Python analysis
- **Visualizations:** matplotlib, SHAP, DCA, calibration curves

Key Features

- **All output in English** (metrics, tables, plot labels)
 - **Built-in statistical safeguards:** proportional hazards testing (Schoenfeld), multicollinearity (VIF), normality tests, homogeneity of variances, calibration (Hosmer-Lemeshow)
 - **Multiple comparison correction** (Bonferroni) in pairwise tests
 - **Stratified survival analysis**
 - **Stepwise predictor selection** with p-value thresholds
 - **DCA** (Decision Curve Analysis) and **calibration curves** for regression models
 - **Bootstrap** confidence intervals for metrics
 - **SHAP** for Random Forest model interpretation
-

2. Installation & Startup

Requirements

Component	Version
Python	≥ 3.11
pip	≥ 21.0

Quick Start (one-click)

macOS:

```
git --version 2>/dev/null || xcode-select --install
git clone https://github.com/dziameshkapavel/PDD_stat.git && cd PDD_stat
./setup_mac.command && ./start_backend.command
```

Windows:

```
git --version >nul 2>&1 || winget install --id Git.Git -e --source winget >nul
git clone https://github.com/dziameshkapavel/PDD_stat.git
cd PDD_stat
setup.bat
start.bat
```

The scripts auto-detect Python 3.11+, create a virtual environment, install dependencies, and start the server.

Manual Setup (alternative)

```
git clone https://github.com/dziameshkapavel/PDD_stat.git && cd PDD_stat
pip install -r app/backend/requirements.txt
cd app/backend
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000
```

Open your browser at: **<http://127.0.0.1:8000>**

3. Project & Data Management

3.1. Creating a Project and Uploading Data

1. Click **" + Create Project "** (left panel).
2. Enter a project name (Latin or Cyrillic).
3. Select a data file:
4. Supported formats: `.xlsx` , `.xls` , `.csv`
5. Click **"Create"**.
6. After upload, the **Cleaning Recommendations** modal appears (see section 3.2).

Re-uploading to an existing project: click the project name with the left mouse button — a file dialog opens.

3.2. Data Cleaning

Upon data upload, the system automatically analyzes every column and suggests a cleaning plan:

Dropped Columns

Rule	Condition	Level
HIGH_MISSING	>30% missing values	Hard
MODERATE_MISSING	20–30% missing values	Soft
CONSTANT	Singleton column	Hard
IDENTIFIER	All unique values (n>10)	Hard
TECHNICAL	Name starts with <code>INFO_</code> , <code>CHECK_</code> , <code>PARAMS_</code> etc.	Hard

Imputation

Remaining columns with missing values: - **Numeric** → filled with **median** - **Categorical/string** → filled with **mode**

Buttons: - **"Apply Cleaning"** — apply recommendations - **"Skip (Keep as is)"** — save raw data as-is

After cleaning, data is persisted as `state/project_data.parquet` (snappy compression).

3.3. Variable Management

Right panel → "Columns" tab:

- **Search** by variable name
- **Type badge:** `num` (numeric), `cat` (categorical), `bin` (binary), `str` (string)
- **Actions:**
 -  — rename
 -  — delete from dataset
 -  — open Label modal:
 - **Chart Name** — display name on plots
 - **Value Labels** — label for values (e.g., `0 = No`, `1 = Yes`)

Quick variable selection: click a variable name to assign it to the active card field (field highlights in color).

3.4. Project Context (Description, Aim)

Click  on a project item → **Project Context:**

- **Description** — project description (used by AI for context)
- **Research Aim** — study objective
- **Notes** — additional notes

Context is automatically injected into the AI assistant's system prompt and AI reports.

4. Interface

4.1. Left Panel — Project Manager

Element	Description
+ Create Project	Create a new project
Project list	Project names. Click → open/upload data.  → context.  → delete
AI Settings	Configure AI assistant
 	Collapse/expand panel

4.2. Workspace

Central area — container for analysis cards (see section 5).

Template menu bar (top):

Menu	Templates
Basic Stats ▼	Categorical, Numeric, Correlation, Spline, Descriptive, Violin, ANOVA, Chart Builder
Regression ▼	Cox, Logistic, LASSO, Random Forest
Survival ▼	Kaplan-Meier
Evaluation ▼	Model Evaluation (Binary), Survival Eval, ROC, Diagnostic Accuracy, Agreement
Prediction ▼	Individual Prediction

Action buttons: - **AI Chat** — open AI assistant chat - **Code** — switch to code editor mode - ☀ / ☾ — toggle light/dark theme

4.3. Right Panel — Variables

Tab	Content
Columns	Variable list with search, type badges, editing
Reports	DOCX report generation
Chain	Analysis history

5. Statistical Analysis Templates

5.1. Categorical Comparison

Method: χ^2 -test / Fisher's Exact Test / Monte Carlo simulation

Parameters: | Field | Description | |-----|-----| | | col1 |
First categorical variable (table rows) | | col2 | Second
categorical variable (table columns) |

Test selection logic: 1. All expected frequencies $\geq 5 \rightarrow \chi^2$ -test
2. Low expected AND 2x2 table $\rightarrow$ **Fisher's Exact Test** 3.
Otherwise $\rightarrow$ **Monte Carlo** (10,000 iterations)

Output metrics:

Metric	Interpretation
chi2	Chi-square statistic
p_value	Significance level; $p < 0.05$ = significant association
df	Degrees of freedom
min_expected_frequency	Minimum expected cell count
all_expected_ge5	All expected ≥ 5 ?
odds_ratio	Odds ratio (Fisher, 2x2 only)

5.2. Numeric Comparison

Method: t-test (independent) / ANOVA or Mann-Whitney U / Kruskal-Wallis H

Parameters: | Field | Description | |-----|-----| | value | Numeric dependent variable | | group | Categorical grouping variable |

Test selection: - If **normal** (Shapiro-Wilk) AND **equal variances** (Levene): - 2 groups → **t-test** + Cohen's d - >2 groups → **ANOVA** + η^2 - Otherwise: - 2 groups → **Mann-Whitney U** - >2 groups → **Kruskal-Wallis H**

Output metrics:

Metric	Interpretation
statistic	Test statistic value
p_value	Significance level
cohens_d	Cohen's d effect size (t-test only): 0.2 = small, 0.5 = medium, 0.8 = large
normality	Shapiro-Wilk test results per group
equal_var	Levene's test result

Plots: Boxplot + barplot (mean $\pm$ SD), 2 panels.

5.3. Correlation Analysis

Method: Pearson (parametric) or Spearman (rank) correlation

Parameters: | Field | Description | |-----|-----| |

`predictors` | Select multiple variables | | `method` | Pearson (assumes normality) or `Spearman` (any distribution) | | `threshold` | | cutoff: 0.3, 0.5 (default), 0.7 |

Output metrics:

Metric	Interpretation
<code>n_variables</code>	Number of variables
<code>n_strong_pairs</code>	Pairs with
<code>strongest_r</code>	Maximum
<code>pairs</code>	Strong pairs with Bonferroni-corrected p-values . Corrects for family-wise error rate (FWER)

p-values are Bonferroni-corrected (multiplied by number of all pairwise tests) to control FWER.

Plot: Correlation heatmap (lower triangle).

5.4. Spline Analysis

Method: Linear spline analysis within Cox or Logistic regression

Parameters: | Field | Description | |-----|-----| |

`variable` | Variable for spline transformation | | `target` | (Logistic) Binary outcome | | `time + event` | (Cox) Time-to-event | | `covariates` | Adjustment covariates | | `knots` | Knot placement: `Median`, `Quartiles`, `Custom` |

Output metrics:

Metric	Interpretation
<code>nonlinear_p</code>	p-value for non-linearity test. p < 0.05 = significant deviation from linearity
<code>knots</code>	Knot positions

Plot: Log-HR / Log-OR vs variable with knot markers + histogram.

5.5. Descriptive Statistics

Method: Summary statistics + Shapiro-Wilk normality test

Parameters: | Field/Option | Description | |-----|-----|
| `covariates` | Variables (empty = all numeric) | | `Include histograms` | Plot histograms per variable |

Output metrics:

Metric	Interpretation
N, Mean, Median, Std, Min, Max, Q1, Q3	Standard descriptive statistics
Skewness	Asymmetry of distribution (0 = symmetric)
Kurtosis	Tail heaviness (3 = normal distribution)
Missing count	Number of NaN values
Outlier count	IQR method: beyond $Q1 - 1.5 \times IQR$ / $Q3 + 1.5 \times IQR$
Shapiro-Wilk W, p	$p < 0.05$ = distribution deviates from normality

5.6. Violin Plots

Method: Violin plots + Mann-Whitney / Kruskal-Wallis test

Parameters: | Field | Description | |-----|-----| | `value` | Numeric variable | | `group` | Grouping variable |

Output metrics:

Metric	Interpretation
<code>mann_whitney_p</code>	Mann-Whitney U p-value (2 groups)
<code>kruskal_p</code>	Kruskal-Wallis H p-value (>2 groups)

Plot: Violin plot, optional box overlay and swarm points.

5.7. ANOVA / Kruskal-Wallis

Method: One-way ANOVA (parametric) or Kruskal-Wallis (non-parametric)

Parameters: | Field | Description | |-----|-----| | value |
Numeric outcome | | group | Categorical grouping (≥ 2 groups)
|

Output metrics (tabs):

Statistics tab:

Metric	Interpretation
test_name	ANOVA or Kruskal-Wallis
F_stat / H_stat	Test statistic
p_value	Significance level
eta_sq	η^2 — proportion of variance explained by group (ANOVA)
omega_sq	ω^2 — less biased effect size than η^2
epsilon_sq	ϵ^2 — effect size for Kruskal-Wallis
trend_test	Jonckheere-Terpstra trend test (≥ 3 groups)

Post-hoc tab:

Metric	Interpretation
Pairwise comparisons	Tukey HSD (ANOVA) or Mann-Whitney with Bonferroni (KW)
significant	$p < 0.05$ after correction

Diagnostics tab:

Metric	Interpretation
all_normal	All groups normal (Shapiro-Wilk $p > 0.05$)
equal_var	Variances equal (Levene $p > 0.05$)

Plot: Boxplot + jitter + barplot (mean $\pm$ SD).

5.8. Chart Builder

Method: Quick visualization (no statistical tests)

Parameters: | Field | Description | |-----|-----| | Chart type | Histogram, Box Plot, Scatter, Bar (mean±SD), Density | | x | X variable | | y | Y variable (scatter only) | | group | Color/group by | | Bins | Histogram bins (default 20) | | KDE curve | Overlay kernel density estimate (histogram) | | Regression line | Regression line with r^2 (scatter) | | Chart Axes | Custom title, axis labels |

Plot: The requested chart type.

5.9. Cox Regression

Method: Cox Proportional Hazards Regression (lifelines CoxPHFitter)

Parameters: | Field | Description | |-----|-----| | event | Event indicator (0 = censored, 1 = event) | | time | Time-to-event | | covariates | Predictors | | Covariate type | Cat / Cont toggle — categorical or continuous | | Reference groups | Reference category for categorical variables | | **Univariate** | Each predictor separately | | **Multivariate** | All predictors jointly | | **Method** | Enter (all at once), Forward (forward selection), Backward (backward elimination) | | p_enter | Entry p-value threshold (default 0.05) | | p_remove | Removal p-value threshold (default 0.10) | | Adjust Survival Curves | Covariate for adjusted survival curves |

Output metrics:

Table tab:

Metric	Interpretation
Beta (β)	Regression coefficient
SE	Standard error
z	z-statistic (β /SE)
HR (Hazard Ratio)	$\exp(\beta)$. >1 = increased hazard, <1 = decreased hazard

Metric	Interpretation
95% CI	95% confidence interval for HR
p-value	Wald test significance

Diagnostics tab:

Metric	Interpretation
C-index	Harrell's Concordance Index — discrimination. 0.5 = random, 1.0 = perfect. Not calibrated for censored data
AIC	Akaike Information Criterion — model quality (lower = better). Not interpretable in isolation
BIC	Bayesian Information Criterion — penalizes complexity more than AIC
Log-likelihood	Log-partial likelihood at convergence
Global LRT p	Likelihood ratio test p-value for overall model significance
Schoenfeld test	Proportional hazards assumption test. p<0.05 = PH violation. Critical diagnostic check
VIF	Variance Inflation Factor. >10 = problematic multicollinearity, >5 = moderate

Steps tab (Forward/Backward only): stepwise selection history.

"Save Risk" button — saves predicted risk (linear predictor) as a new column.

Plots (checkboxes, all optional): - **Forest plot** — HR with 95% CI - **Survival curves** — adjusted survival curves - **Residuals** — Martingale residuals (linearity diagnostic)

Assumption checks: - EPV (Events Per Variable) ≥ 10 — warning if <10 - Schoenfeld — PH test per variable and globally - VIF — multicollinearity - **Stepwise warning:** p-values do not account for the selection process and may be inflated

5.10. Logistic Regression

Method: Logistic Regression with L2 penalty,

`class_weight='balanced'`

Parameters: | Field | Description | |-----|-----| | `target` |

Binary target variable | | `predictors` | Predictors | | Covariate type | Cat / Cont toggle | | `Reference groups` | Reference category | | **Univariate** | Each predictor separately | |

Multivariate | All predictors jointly | | **Method** | Enter ,

Forward , Backward | | **Validation** | None , Train/Test (80/20) , `Cross-val` (5-fold CV) | | **Plots** | Checkboxes: ROC, DCA, Calibration. All optional, select what you need |

Output metrics:

Table tab:

Metric	Interpretation
Beta (β)	Regression coefficient
OR (Odds Ratio)	$\exp(\beta)$. >1 = higher odds, <1 = lower odds
95% CI	95% confidence interval for OR
p-value	Wald test significance

Diagnostics tab:

Metric	Interpretation
AUC	Area Under ROC Curve — discrimination. 0.5 = random, 0.8 = excellent, 1.0 = perfect
Accuracy	$(TP+TN)/(Total)$ — overall correct classification
Sensitivity	$TP/(TP+FN)$ — true positive rate
Specificity	$TN/(TN+FP)$ — true negative rate
Hosmer-Lemeshow	Calibration test: χ^2 , df, p-value. $p < 0.05$ = significant miscalibration
VIF	Multicollinearity (>10 = problematic)
Warnings	VIF errors, convergence issues

Steps tab (Forward/Backward): stepwise selection history.

"Save Probabilities" button — saves predicted probabilities as a new column.

Plots (checkboxes, all optional): - ROC curve - DCA (Decision Curve Analysis) with Treat All / Treat None baselines - Calibration curve with Brier Score

Assumption checks: - Target must be binary (exactly 2 unique values) - Minimum 20 complete cases - All predictors standardized (StandardScaler) - **Warning** when validation="none": metrics are optimistic

5.11. Logistic LASSO

Method: L1-regularized logistic regression (LASSO)

Parameters: | Field | Description | |-----|-----| | `target` | Binary target | | `covariates` | Predictors (empty = all numeric + encoded categorical) | | `Auto-select` | Automatic C selection via 5-fold CV | | `C` | Inverse regularization strength (if auto-C off). Smaller C = stronger regularization | | **Plots** | Checkboxes: Coef Plot, CV Curve, DCA, Calibration. All optional |

Output metrics:

Metric	Interpretation
AUC	ROC AUC (training data — optimistic)
Accuracy	Overall accuracy (training — optimistic)
<code>n_features_selected</code>	Features with non-zero coefficient
<code>n_features_zeroed</code>	Features zeroed out (removed by LASSO)
<code>best_C</code>	Best C value (if auto-select)
<code>selected_features</code>	List: feature, OR, coefficient
<code>intercept</code>	Model intercept

Plots (checkboxes, all optional): - Coefficient bar plot (selected + top 30 all features) - CV curve (if auto-select C) - DCA (Decision Curve Analysis) - Calibration curve

Important: - Metrics are on **training data** — **optimistic** for generalization - `class_weight='balanced'` — handles class imbalance - `solver='saga'` — appropriate for L1 penalty

5.12. Random Forest

Method: Random Forest Classifier (sklearn, 200 trees, `max_depth=10`, `min_samples_leaf=5`, `class_weight='balanced'`)

Parameters: | Field | Description | |-----|-----| | `target` | Target variable (binary or multiclass) | | `exclusions` | Columns to exclude (comma-separated) | | `Top features` | 10, 15 (default), 20, All | | `SHAP analysis` | Enable SHAP calculation (may be slow on large datasets) |

Output metrics:

Metric	Interpretation
OOB score	Out-of-Bag R^2 — unbiased performance estimate from unused trees
Accuracy	Test set accuracy (75/25 split)
AUC	ROC AUC (binary targets only)
Precision	$PPV = TP/(TP+FP)$
Recall	$Sensitivity = TP/(TP+FN)$
F1	Harmonic mean of precision and recall
<code>top_features</code>	Gini importance — contribution of each feature to node purity. Not a p-value!
<code>shap_features</code>	Mean absolute SHAP — more interpretable feature contribution magnitude

Plots: - Feature importance bar (Gini) - SHAP bar + summary plots (if enabled)

Post-run buttons: - **"Save Model"** — saves model as .joblib (with metadata: features, target, metrics) - **"Save**

Probabilities" — saves predicted probabilities as new column

Note: Auto-excludes columns matching: `_event`, `event_`, `_time`, `time_`, `predicted_`, `_prob`, `censored`, `followup`, `cox_`, `logistic_`. Only numeric predictors are used.

5.13. Kaplan-Meier

Method: Kaplan-Meier survival curves / Nelson-Aalen cumulative hazard

Parameters: | Field | Description | |-----|-----| | `time` | Time-to-event | | `event` | Event indicator | | `group` | Grouping variable | | `stratify` | Stratification variable (separate analysis per stratum) | | `Survival` / `Hazard` | Plot type: survival probability or cumulative hazard | | `95% CI` | Show confidence intervals | | `Chart Axes` | X step (months), Y step (%), labels |

Output metrics:

Metric	Interpretation
<code>summary</code>	Per group: n, events, median survival (months)
Log-rank p	Overall log-rank test p-value. p<0.05 = groups differ significantly
Pairwise log-rank	Bonferroni-corrected pairwise comparisons. Shows which specific groups differ
<code>number_at_risk</code>	Number-at-risk table at key time points
<code>survival_probability</code>	S(t) survival probability table
<code>median_survival</code>	Median survival per group. <code>NR</code> = not reached

Plot: Kaplan-Meier survival curve S(t) or Nelson-Aalen cumulative hazard H(t).

5.14. ROC Analysis

Method: ROC analysis for multiple numeric predictors vs binary target

Parameters: | Field | Description | |-----|-----| | `target` | Binary target | | `Event value` | Value considered "event" (1 or 0) | | `predictors` | Numeric predictors |

Output metrics:

ROC Results tab:

Metric	Interpretation
AUC (95% CI)	Area Under ROC Curve. Hanley-McNeil CI method
Threshold	Optimal cutpoint via Youden's J (sens + spec - 1)
Sensitivity	True positive rate at optimal threshold
Specificity	True negative rate at optimal threshold
PPV	Positive predictive value
NPV	Negative predictive value

Comparison tab (≥2 predictors):

Metric	Interpretation
DeLong test	Pairwise AUC comparison. p<0.05 = AUCs differ significantly

Plot: Combined ROC curves for all predictors.

5.15. Model Evaluation (Binary)

Method: Comprehensive binary classification evaluation with bootstrap CI

Parameters: | Field | Description | |-----|-----| | target | Binary target | | predictors | Predicted probability columns (0-1) | | Bootstrap samples | 500, 1000 (default), 2000 | | DCA | Include Decision Curve Analysis | | Calibration | Include calibration curves |

Output metrics:

Metric	Interpretation
AUC (bootstrap CI)	ROC AUC with bootstrap 95% CI
Accuracy	Overall classification accuracy
Sensitivity / Specificity	True positive and negative rates

Metric	Interpretation
PPV / NPV	Predictive values
F1	Harmonic mean of precision and recall
Brier Score	Mean squared probability error. 0 = perfect, <0.25 = good, >0.25 = poor
DCA Net Benefit	Clinical net benefit at thresholds 5-50%. $NB = (TP/n) - (FP/n) \times (p/(1-p))$

Plots: - DCA curves (all models on one plot) - Calibration curves (one per model)

5.16. Survival Prediction Evaluation

Method: Survival model evaluation: C-index, time-dependent AUC, bootstrap CI

Parameters: | Field | Description | |-----|-----| | `time` | Time-to-event | | `event` | Event indicator | | `predictors` | Risk score/prediction columns | | `Evaluation time points` | Time points for AUC (comma-separated, e.g., "12,24,48") | | `stratify` | Stratification variable | | `Bootstrap samples` | 500, 1000 (default), 2000 | | `Smooth` | Smoothed AUC curves |

Output metrics:

Metric	Interpretation
C-index (bootstrap CI)	Harrell's C-index with bootstrap 95% CI. 0.5 = random
Time-dependent AUC	Cumulative dynamic AUC as function of time
ΔC (model comparison)	C-index difference between models with bootstrap p-value
Strata	Separate results per stratum level

Plot: Time-dependent AUC (optionally smoothed).

5.17. Diagnostic Accuracy

Method: Diagnostic accuracy metrics for binary tests vs gold standard

Parameters: | Field | Description | |-----|-----| | target |
Reference standard (binary, "gold standard") | | predictors |
Index tests (binary or probability 0-1) |

Output metrics (per test):

Metric	Interpretation
Sensitivity (95% CI)	TP/(TP+FN). Proportion of correctly identified positives
Specificity (95% CI)	TN/(TN+FP). Proportion of correctly identified negatives
PPV (95% CI)	TP/(TP+FP). Probability that a positive test reflects true disease
NPV (95% CI)	TN/(TN+FN). Probability that a negative test reflects true absence
Accuracy (95% CI)	(TP+TN)/(Total). Overall correct classification
LR+ (95% CI)	Positive likelihood ratio = Sens/(1-Spec). >10 = strong evidence
LR- (95% CI)	Negative likelihood ratio = (1-Sens)/Spec. <0.1 = strong evidence
DOR	Diagnostic Odds Ratio = (TP×TN)/(FP×FN). 1 = useless test

Comparison tab (>1 test):

Metric	Interpretation
McNemar test	Paired comparison. p<0.05 = tests differ significantly

Contingency tables tab: TP/FP/FN/TN per test.

Plot: Forest plot of sensitivity and specificity with 95% CI.

Note: Probability columns (0-1) are auto-thresholded at 0.5.

5.18. Agreement Analysis

Method: Cohen's Kappa (2 raters) / Weighted Kappa (ordinal) / Fleiss' Kappa (3+ raters)

Parameters: | Field | Description | |-----|-----| |
predictors | Rater columns (≥ 2 required) | | Nominal /
Ordinal | Scale type | | Linear / Quadratic | Weight type
(ordinal only) |

Output metrics:

Metric	Interpretation
Kappa	Agreement coefficient corrected for chance. Range: -1 to 1
95% CI	Confidence interval
p-value	Significance vs zero
Interpretation	Landis & Koch: Poor (≤ 0), Slight (0.01-0.20), Fair (0.21-0.40), Moderate (0.41-0.60), Substantial (0.61-0.80), Almost Perfect (0.81-1.00)
percent_agreement	Raw agreement percentage

For ≥ 3 raters: Fleiss' Kappa with per-category breakdown.

Plot: Confusion matrix heatmap (2 raters).

5.19. Individual Prediction

Method: Single-patient prediction from saved model with SHAP explanation

Parameters: | Field | Description | |-----|-----| | **Select File** | Upload .joblib or .pkl model | | Patient form | Radio buttons for categorical features, number inputs for numeric |

Output metrics:

Metric	Interpretation
prediction	Predicted class
probability	Probability of class 1

Plot: SHAP waterfall plot — contribution of each feature to the prediction.

6. AI Chat

6.1. AI Settings

Click **"AI Settings"** (left panel or above AI Chat):

Parameter	Description
Provider	Ollama (Local) — run models locally. Groq (Cloud) — cloud API
Ollama URL	Ollama server address (default <code>http://localhost:11434</code>)
Ollama Model	Model (loaded after "Test Connection")
Groq API Key	API key (get at console.groq.com)
Groq Model	Model (list loaded after "Test Connection")
Temperature	Generation temperature: 0 = deterministic, 1 = creative
Max Tokens	Maximum response length (100-8000)
System Prompt	AI instruction prompt (includes project context, dataset info, up to 5 most recent analyses with metrics and output preview)

Buttons: - **"Test Connection"** — test provider connectivity and load model list - **"Connect"** — confirm provider selection - **"Reset to default"** — reset system prompt to default - **Save / Cancel** — save or discard settings

6.2. Modes: Assistant and Coder

AI Assistant (default): - Conversational chat with AI - Answers statistical questions, interprets results - Has context of project description, dataset, and analysis history - Supports markdown formatting

AI Coder ("Coder" toggle): - Generates Python analysis code
- Auto-extracts `python` code blocks and executes them -
Shows code output and generated plots - Self-corrects on
errors (up to 3 retries) - Uses temperature 0 (deterministic
output)

7. Reports

Right panel → **Reports** tab.

7.1. Standard DOCX Report

Parameter	Description
Analyses	All analyses or Selected (with checkboxes)
Sections	Dataset overview, Results tables, Key metrics
Title	Report title
Generate DOCX	Create report. Download via link

Contents: - Dataset overview (row count, column types) - For each analysis: metrics, tables from preview text

7.2. AI Report

Parameter	Description
Language	Report language (user prompt, default "English")
Generate AI Report	Create AI-generated report

Contents (AI-generated): 1. **Objective** — based on project context 2. **Materials** — dataset description + basic stats 3. **Methods** — statistical methods used 4. **Results** — key findings with exact numbers (HR, OR, p-values, AUC, C-index) 5. **Conclusion** — brief summary

Requirements: - AI must be configured (valid provider and credentials) - At least one analysis in history (max 5 included in prompt)

8. Code Editor Mode

Click **"Code"** in the header.

Action	Description
Type Python code in the monospace text area	
Run	Execute via <code>exec()</code> on the server
Clear Code	Clear editor
Clear History & Plots	Delete all analysis history and plot files
Exit Code Mode	Return to analysis view

Available variables in code: - `df` — project pandas DataFrame - `pd`, `np`, `plt`, `shap` — standard libraries - `save_plot(name)` — save plot with auto-close - `fmt_p(p)` — format p-value (shows `<0.0001` if tiny) - `get_label(var, value)` — variable/value labels

9. Analysis History (Chain)

Right panel → **Chain** tab.

- Chronological list of completed analyses (reverse order, max 200 entries)
 - Color coding:
 - **Blue** — regression
 - **Green** — survival
 - **Amber** — evaluation
 - **Gray** — basic tests
 - **Orange** — LASSO
 - **Purple** — Random Forest / Random Survival Forest
 - Click an item to restore the analysis card with results
-

10. Tips & Best Practices

Action	Tip
Enter in AI input	Send message
Shift+Enter	New line
Click a variable	Assign to active field
Hide Charts	Free up memory by hiding plots
☀ / ☾	Toggle theme

Workflow recommendations

1. **Start with Descriptive Statistics** — inspect distributions, missing values, outliers
2. **Use Categorical + Numeric Comparison** for preliminary screening
3. **For regression:** start with Univariate, then Multivariate (Enter), then Stepwise if needed
4. **Always check Schoenfeld PH test** in Cox regression — this is the key assumption
5. **VIF > 10** — remove collinear predictors and re-run
6. **DCA** helps assess clinical utility, not just statistical significance
7. **Bootstrap CI** provides more realistic metric precision estimates
8. **Don't rely solely on Stepwise p-values** — they don't account for the selection process
9. **LASSO with auto-select C** is a good choice for high-dimensional feature selection
10. **AI Coder mode** is useful for quick ad-hoc analyses without leaving the app